MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2022/23 END OF SEMESTER EXAMINATIONS

PHYS-1101: MECHANICS AND THERMAL PROPERTIES OF MATTER

[100 marks]

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DATE: 8th May 2023 **TIME:** 3hrs: 9:00 - 12:00

INSTRUCTIONS

1. Write your **Registration Number** on each page of your answer shee.
2. Start each question on a **NEW** page.
3. This paper contains **7 printed pages** and **6 QUESTIONS**, please check.
4. Answer **ALL** questions.
5. Show your work clearly.
6. Marks allocated for each question are indicated against it.
7. Round off your final answer to 2 decimal places.

USEFUL PHYSICAL CONSTANT(S)

Acceleration due to gravity	g	$9.8 m/s^2$
Gravitation constant	G	$6.67x10^{-11} N.m^2/kg^2$
Radius of the Earth	R_E	$6.37x10^6 m$
Mass of the Earth	M_E	$5.98x10^{24} kg$
Latent Heat of fusion of water	L_w	6.0×10^3 J.kg^{-1}
Boltzmann's constant	k	1.381×10^{-23} J K^{-1}
Gas constant	R	8.315 J mol^{-1} K^{-1}
Pressure of a standard atmosphere		1.013×10^5 Pa (or N m^{-2})
Speed of light (in vacuum)	c	2.998×10^8 m s^{-1}
Speed of sound in air	v_a	343 m/s
Speed of sound in water	v_w	1480 m/s
Specific heat capacity of water	c_w	4.184×10^3 Jkg^{-1} K^{-1}
Latent heat of fusion of ice	L_{ice}	334000 Jkg^{-1}
Specific heat capacity of glass	c_g	8.37×10^2 Jkg^{-1} K^{-1}
Specific heat capacity of steam	c_s	2.01×10^3 Jkg^{-1} K^{-1}
Latent heat of vaporisation of water,	L_w	2.260×10^6 Jkg^{-1} K^{-1}
Specific heat capacity of ice	c_{ice}	2200 Jkg^{-1} K^{-1}
Room temperature	T_R	25 ^{0}C
Coefficient of linear expansion of Lead.		2.9×10^{-5} K^{-1}
Coefficient of linear expansion of steel.		1.70×10^{-5} K^{-1}
S.T.P		273 K and 1.01×10^5 Pa

Question 1.

A. A car uniformly accelerates from rest to a velocity of 50 km/hr after covering a displacement of 100 km eastwards. What is its acceleration in SI units? **[5 marks]**

B. The amount of heat (Q) gained or lost by a sample can be calculated using the equation $Q = mc\Delta T + \beta d$ and it is dimensionally correct. Where m is the mass of the sample, c is the specific heat, d is the displacement and ΔT is the temperature change. What are the dimensions of β and what quantity does it represent? **[4 marks]**

C. The distance between a town and a factory is 30 km. A man started to walk from the factory to town at 6:30 am while the cyclist left the town for the factory at 6:40 am riding at a constant speed of 18 km/h. The man met the cyclist after walking 6 km. At what time did they meet? **[4 marks]**

D. The variation of velocity of a particle with time moving along a straight line is illustrated in **figure 1**. What is the distance travelled by the particle in four seconds? **[6 marks]**

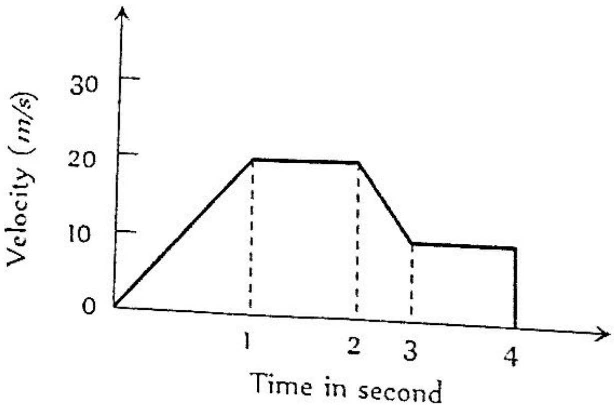

Figure 1

Question 2.

A. A ship leaves Port A and sails 61.1 km on Course 131° to Port B, then Sails 76.5 km course 36° to Port C. Find the displacement covered by the ship from port A to port C **[5 marks]**

B. A particle is projected from ground with a velocity of 20 m/s at an angle of 30° with the horizontal. Determine the time(s) at which the vertical displacement of the projectile is 3 m. **[6 marks]**

C. A bullet of mass 0.005 kg moving with a speed of 200 m/s enters a heavy wooden block and it is stopped after a distance of 50 cm. What is the force exerted by the block on the bullet? **[4 marks]**

D. Two crates, 10 kg and 15 kg are connected with a thick rope as in **figure 2.** A force of 500 N is applied to the right. The boxes move with an acceleration of 2 m/s^2 to the right. One third of the total frictional force is acting on the 10 kg block and two thirds on the 15 kg block. Calculate the tension in the rope. **[5 marks]**

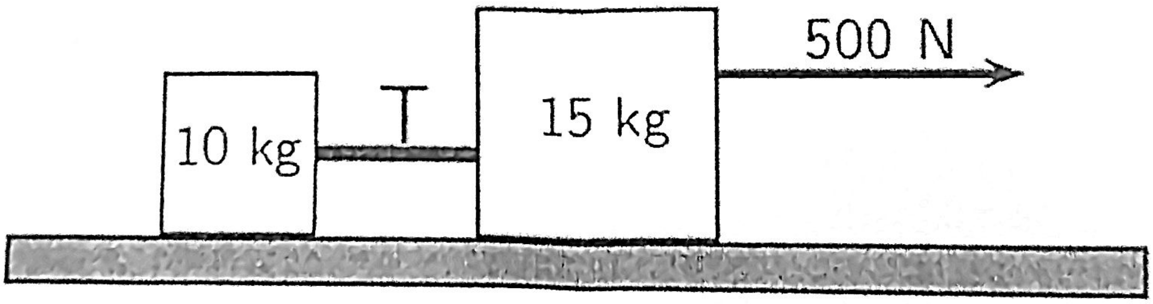

Figure 2

Question 3.

A. Define work and mechanical energy and explain how they are related to each other. **[6 marks]**

B. A 10.0-kg block A is connected to a 10-kg block B by a cable as **Figure 3** Shows. The inextensible cable and pulley are massless and frictionless. The coefficient of kinetic friction between block A and the inclined surface is 0.300. A 200-N force is applied to block A in the direction shown, and pulls block B upward by 1 m. How much work is done by gravity on block A?

[3 marks]

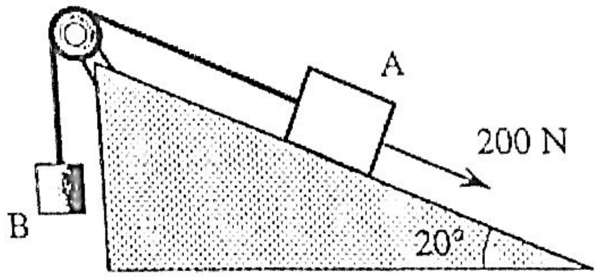

Figure 3.

C. A block of mass 4 kg is dragged 2 m along a horizontal surface by a force F = 30 N acting at 53^0 to the horizontal. The initial speed is 3 m/s and if the coefficient of kinetic friction is 0.125. Find the change in kinetic energy of the block and its final speed. **[8 marks]**

Question 4

A. State the law of conservation of linear momentum? **[2 marks]**

B. An object of mass 3 kg is moving along the x-direction, with a speed of v=10 m/s and collides with another object of mass 2kg moving in the y-direction, with speed of 5 m/s. The two objects collide and stick to each other. What is the velocity of the combined object after collision? **[5 marks]**

C. The wheel of a car has a radius of 0.29 m and it is being rotated at 830 rpm on a tyre-balancing machine. Determine the speed at which the outer edge of the wheel is moving. **[3 marks]**

D. The maximum tension that a 0.60 m string can tolerate is 20 N. A 0.45 kg ball attached to this string is whirled in a vertical circle. What is the maximum speed of the ball at the top of the circle? **[4 marks]**

Question 5.

A. State the zeroth law of thermodynamics. **[2 marks]**

B. Explain the process of calibrating a Celsius scale. **[4 marks]**

C. If MUST want to come up with its own temperature scale to be exclusively used in Malawi. What would be the conversion formula with the Celsius scale, if its boiling point is at 200ºM and freezing point is at 40ºM? **[4 marks]**

Question 6.

A. Define heat capacity

[2 marks]

B. i) How much energy would it take to raise the temperature of 500 grams of water in a kettle from 21°C to 100°C?

[2 marks]

ii) The kettle has a power rating of 3kW. How long would the process in (i) take?

[2 marks]

C. How much heat must be removed from 1.5 kg of water at 0ºC to change its state to ice at 0ºC?

[2 marks]

D. Show that the coefficient of volume expansion (β) is three times the coefficient of linear expansion α, that is, $\beta=3\alpha$.

[7 marks]

E. A solid plate of Lead is 8 cm x 12 cm at 15°C. What is the change in area and the new area of the Lead plate if the temperature increases to 95°C?

[5 marks]

END OF QUESTIONS